

CRAD Process, Role Navigation, and Security Features

Center for Research and Development (CRAD) · School Management System 2

Document generated from the running codebase (sidebar, authentication, CRAD modules) · 23 September 2026

This packet maps (1) the end-to-end CRAD research workflow, (2) every sidebar / navbar function per user role, and (3) security controls with where they appear in the system and the code that implements them.

1. Contents

1. CRAD academic pipeline
2. Official title / group rule
3. Grant funding pipeline (parallel track)
4. Actors and responsibilities
5. Shared top navbar
6. Sidebar functions by role
7. Security features and proof
8. Limitation to disclose

2. CRAD academic pipeline

Capstone / thesis work is a gated chain. Later steps are blocked until earlier records exist (registered group, assigned adviser, approved chapters, scheduled defense).

1. **Title proposal.** Student submits CRAD Form S2 (Title Approval) in Student Portal → Research Proposal (`modules/student-portal/pages/research-proposal-submission.php`). If the title was Returned, the sidebar opens the returned-title notification instead of a blank form.
2. **Triple sign-off.** Adviser, Research Coordinator, and CRAD Officer must all approve, with stored signature data. See Section 3.
3. **Register proposal.** CRAD Officer registers the title (`registration_status` : Pending → Registered) in Register Proposal.
4. **Research group number.** Official group number is issued (`research-group-number.php`).
5. **Assign adviser.** Department Head retrieves approved research, checks availability, and assigns an adviser.
6. **Research development.** Student uses My Research, Research Plan, Milestones, Progress Updates, Adviser Feedback. Adviser monitors the same group from the faculty workspace.
7. **Submit Chapters 1–3.** Student uploads documents. Grammarian scores five criteria at 20% each (content, methodology, references, format, grammar).

8. **Pre-oral defense scheduling.** Research Director (or operational Admin) queues groups that have approved Chapters 1–3, proposes slots (max 3 panel members), and finalizes venue/time.
9. **Panel evaluation.** Assigned panel members score the defense (Evaluation & Scoring).
10. **Implementation / Chapters 4–5.** Student continues milestones (system development, testing, documentation) in 2nd semester phase.
11. **Final manuscript.** Student uploads versions; CRAD / adviser review and revision compliance.
12. **Final defense.** Director schedules Final Defense; panel uses Final Defense Evaluation.
13. **Revisions.** Adviser tracks Final Defense Revisions / Revision Monitoring.
14. **Research clearance.** Student uploads collage/college payment and the signed clearance image. CRAD Officer approves or rejects from the clearance inbox.
15. **Repository / Publications & IP.** Completed work is recorded and archived.

3. Official title / group rule (integrity gate)

A research group is treated as official only when all of the following are true on `crad_title_approvals` :

- `status = 'Approved'`
- `coordinator_status = 'Approved'`
- `crad_status = 'Approved'`
- Adviser, coordinator, and CRAD signature fields are non-empty

Implemented by `cradValidTitleApprovalWhereSql()` in `modules/crad/config/config.php` . Department Head “Retrieve Approved Research” and registry queries reuse this rule.

Stage	Key statuses	Where it appears
Title (Form S2)	Pending / Approved / Returned; <code>coordinator_status</code> ; <code>crad_status</code>	Student Research Proposal; CRAD Register Proposal
Registration	Pending → Registered + <code>proposal_number</code>	<code>register-proposal.php</code> , <code>research-group-number.php</code>
Adviser / panel	Assigned / Confirmed; availability Available	Department Head Research Management
Chapters 1–3	Submitted → scored / For Revision; all three approved	Student Submit Chapter 1-3; Grammarian For Evaluation
Milestones	Not Started / in progress / Approved / Completed; 1st vs 2nd semester	Student Milestones; Adviser Research Progress
Manuscript	Submitted / Under Review / For Revision / Approved	Final manuscript pages
Clearance	Uploaded signed image → Approve / Reject	Student clearance pages; CRAD inbox
Grant	<code>workflow_status</code> ; Returned loops to researcher	Approval Workflows YES/NO + signature dialog

4. Grant funding pipeline (parallel track)

Researchers (student or adviser) apply from Grant Opportunities. Scoring happens in Reviewer Evaluation. Funding then requires six sequential office signatures. A step cannot be skipped. Choosing NO sets status Returned; the applicant sees it under Revisions Requested and the chain restarts.

Order	Office	role_key	Function
1	Academic Adviser	adviser	First signature on the grant application
2	Department Chair	department_chair	College/department sign-off
3	College Dean	hr	Dean step (Dean account uses faculty HR workspace)
4	Research Office	research_office	Research office review
5	VPAA	vpaa	VPAA sign-off (Accreditation shell URLs)
6	Finance Office	finance	Final sign-off; can release disbursement

Code: `grantApprovalStepDefinitions()` in `modules/crad/includes/grant-approval-helpers.php`.

5. Actors and responsibilities

Actor	Function in CRAD
Student	Submit title, chapters, progress, manuscript, clearance files, grant applications
Adviser	Sign title, monitor group, grant step 1, feedback on updates
Research Coordinator	View approved titles; coordinator_status + signature on Form S2
CRAD Officer	crad_status + signature; register proposal; group number; approve clearance
Department Head	Assign adviser, then panel when defense-ready
Grammarian	Score Chapters 1–3 before pre-oral queue
Research Director / sms_admin	Schedule pre-oral and final defense; venues; finalize
Panel / Dept. Chair	Score assigned defenses; chair also grant step 2
Dean, Research Office, VPAA, Finance	Grant chain steps 3–6; finance disbursement
Super Admin	Users, RBAC, logs, settings — excluded from operational CRAD

6. Shared top navbar

Source: `includes/navbar.php`. Same chrome for every signed-in user; notification payload is role-specific.

Control	Function
Sidebar toggle	Collapse / expand left navigation
Brand / logo	Goes to that role's home URL (<code>smsRoleHomeUrl()</code>)

Control	Function
Global search (Ctrl+K)	Searches only modules and pages the role can see
PH clock	Philippine Standard Time (Asia/Manila)
Theme toggle	Light / dark
Messages	Inbox dropdown
Notifications	Returned titles, assignments, defenses
User menu	My Profile, Security, Logout (confirm modal)

Sidebar templates come from `smsSidebarMode()` in `includes/navigation-context.php` : *student*, *faculty_workspace*, or *admin_modules*.

7. Sidebar functions by role

Source of truth: `includes/sidebar.php` plus `getVisibleModules()` in `includes/authentication.php` and `$MODULES` in `config/config.php`.

7.1 Student · home: student-portal dashboard

Nav group	Item	Function
Overview	Dashboard	Student home: group, payments, research status
Student Information	My Profile / Student ID	Profile and digital ID
Financial	Account Balance / Payment History	Balances and paid transactions
Academics	Class Schedule, Academic Records, Subject & Professors, Grades Portal	Academic self-service
Research	Research Proposal	Form S2 title approval (or returned-title notice)
Research Development	My Research, Research Plan, Milestones, Progress Updates, Adviser Feedback, Final Manuscript	Implementation and monitoring after group exists
Document Submission	Submit Chapter 1-3, My Submissions, Submission Status, Submission History	Chapter upload and version tracking (CSRF + file whitelist)
Research Clearance	Upload Collage Payment / Research Services Clearance	Payment proof and signed clearance image
Core System	Grant Opportunities, Proposals & Applications, Revisions Requested	Apply to published grant calls
Funded Research / Outputs	Funded Research, Project Milestones, Publications & IP	Execute funded work and record outputs
System	Security Settings	Password, OTP, passkey, module logs

7.2 Adviser · home: assigned-research.php

- **Approved Research** — View Approved Research
- **Research Monitoring** — My Research Groups, Final Defense Revisions, Research Progress, Milestones, Revision Monitoring, Submitted Updates, Adviser Feedback
- **Grades Portal / Schedule / Profile** — Grade Entry, Records, Summary; My Schedule, Defense Schedule; Profile, Expertise, Availability (used in matching)
- **Grants** — Opportunities, Applications, Revisions, Funded Research, Milestones, Reviewer Evaluation, Approval Workflows (grant step 1)
- **Hidden for adviser only** — the “My Research” submenu (Assigned Research / Final Manuscript Review) is unset in the sidebar; backend pages are not deleted

7.3 Panel · home: assigned-defenses.php

- **DEFENSE** — Assigned Defenses, Defense Details, Evaluation & Scoring, Evaluation History, Final Defense Evaluation
- **PROFILE** — My Profile, Availability (used when Department Head checks panel load)
- **System** — Security Settings

7.4 Grammarian · home: for-evaluation.php

- **Evaluation** — For Evaluation (queue), Evaluation & Scoring (5×20% rubric), Evaluation History
- **System** — Security Settings

7.5 Research Director · home: research-director.php

The SMS left sidebar only lists Security Settings. Operations live in the in-page views of

`modules/faculty/pages/research-director.php` :

- Overview
- Ready for Scheduling (pre-oral and final)
- AI Scheduling Optimizer, Proposed Schedules, Alternative Time Slots
- Calendar, Venues, Finalize Schedule
- Researchers, Advisers, Panel Members
- Notifications, Defense Results, Digital Scores, Defense History
- Proceed to Archiving

Access is gated by `smsCanManageDefenseScheduling()` (granted admin or research_director).

7.6 Research Coordinator · home: crad/index.php

`smsResearchCoordinatorCradModule()` trims CRAD to a single item: **View Approved Research**, plus System → Security Settings.

7.7 Department Head · home: research-coordinator-management.php

- Research Coordinator Management (ungrouped)
- **A. Adviser Assignment** — Retrieve Approved Research, Find/Contact Adviser, Check Adviser Availability, Assign Research Adviser, View/Manage Assignments
- **B. Panel Assignment** — Retrieve Defense-Ready Research, Select Panel Members, Check Panel Availability, Assign Panel Members

Direct URLs outside `smsDepartmentHeadWorkflowPaths()` are blocked.

7.8 CRAD Officer · home: dashboard

- **Research Clearance** — Approve Signed Clearance (inbox), Approval Clearance Payment
- **Research Proposal** — Register Proposal, Research Group Number
- **Research Management** — Capstone Group/Student Registry (defense scheduling and reviewer-evaluation are skipped in this role's CRAD nav)
- **Documents / Reports** — manuscript review/approval/revision, publication create, repository, analytics reports

`research-coordinator-management` , grant dashboard, and grant opportunity slugs are removed from this role's visible CRAD module.

7.9 Operational Admin (sms_admin)

`getVisibleModules()` replaces every module with a CRAD-only workspace:

- Approval Clearance Payment, Dashboard & Analytics, Grant Opportunities, Proposals & Applications
- Full pre-oral and final defense scheduling (same director views)
- Announcements (granted-admin extra)
- Security via Profile → Security tab

7.10 Super Admin · User Management only

- User Accounts — create, lock, assign roles
- Role & Permissions — `sms2_role_permissions` matrix
- Module Security — password-reset approvals, per-module logs
- Activity Logs — `sms2_activity_logs`
- System Settings — idle timeout, CAPTCHA/Turnstile, maintenance

Hard rule: Super Admin is stripped of `crad` and `crad_grant` in `smsAllowedModuleKeysForRole()` .

7.11 Grant-chain offices

Role	Sidebar	Function
Department Chair	DEFENSE (same as panel) + Review & Workflow. CRAD module is unset.	Panel scoring plus grant step 2
Research Office	Review & Workflow only (CRAD unset)	Grant step 4
Dean / HR (hr)	Faculty Management + Review & Workflow	HR faculty ops plus grant step 3
VPAA	Review & Workflow under Accreditation URLs (accreditation module hidden)	Grant step 5
Finance	Full Payment module + Approval Workflows (reviewer-evaluation skipped)	Grant step 6; budget disbursement is finance-gated
Research Grant officer	<code>crad_grant</code> : publish calls, applications, evaluation, funding, monitoring, outputs	Operate the grant program
Review Committee	Reviewer Evaluation + Security only	Score applications

7.12 Other office defaults (smsDefaultModulesForRole)

- **admission** — Enrollment Management (pre-registration through enrollment dashboard)
- **registrar** — Registrar + Curriculum + Scheduling
- **qa** — Accreditation Management, plus Review & Workflow for grants
- **it_office** — Online Learning & LMS
- **osa** — Co-Curricular (clubs, events, volunteer hours)

Live installs may add modules through Role & Permissions; the table above is the code fallback when the matrix is empty.

8. Security features and proof

For defense: show the UI first, then name the file. Every row exists in this repository.

Feature	How it is seen in the running system	Code proof
Login required	Opening a module while logged out redirects to <code>login/login.php</code> . AJAX/API calls receive JSON 401 <code>session_expired</code> instead of HTML (avoids fake "success").	<code>requireAuth()</code> in <code>includes/authentication.php</code>
Role-based modules (RBAC)	Student never sees User Management. Super Admin never sees CRAD. Sidebar is built from allowed modules only.	<code>smsAllowedModuleKeysForRole()</code> + table <code>sms2_role_permissions</code> ; students forced to <code>student_portal</code> only
Page-level role gates	Pasting another role's URL shows Forbidden or a redirect, not the page.	Examples: <code>submit-chapters.php</code> requires student; CRAD clearance uses <code>rsccanManageAsCrad()</code> ; director uses <code>smsCanManageDefenseScheduling()</code> ; grammarian uses <code>chapterIsEvaluator()</code>
Password hashing	Sign-in form. Failed login does not reveal whether the username exists (dummy <code>password_verify</code> on unknown user).	<code>password_verify</code> against <code>password_hash</code> ; dummy hash path in <code>authentication.php</code>
CAPTCHA / Cloudflare Turnstile	Checkbox or Turnstile widget on Sign In, evaluated before password and 2FA.	<code>includes/captcha.php</code> ; <code>login.php</code> : CAPTCHA first so bots never reach password check
Passkey (WebAuthn)	"Sign in with passkey" on login; enroll under Security Settings.	<code>includes/passkey.php</code> , <code>assets/js/passkey.js</code> , <code>api/passkey.php</code> , table <code>sms2_user_passkeys</code>
Authenticator TOTP + email OTP	Security Settings / Profile: enable Google Authenticator; 6-digit codes. Password change requires Authenticator or email OTP.	<code>includes/totp.php</code> ; <code>smsCreateOtp()</code> / <code>smsVerifyOtp()</code> ; <code>sms2_security_otp</code> stores SHA-256 of the code
Login throttle / logout	After too many failures, Sign In is disabled until cooldown; wait time is shown.	<code>sms2_login_throttles</code> ; <code>smsLoginGateSet()</code> / <code>smsLoginGateStatus()</code>
Force password change	After an admin reset, next login goes to Change Password.	<code>must_change_password</code> column and session flag → <code>login/change-password.php</code>

Feature	How it is seen in the running system	Code proof
CSRF tokens	Hidden <code>csrf_token</code> in forms. Invalid token shows "Security check failed."	<code>csrfToken()</code> / <code>csrfVerify()</code> using <code>hash_equals</code> in <code>includes/security.php</code>
Session hardening + idle logout	Cookie name <code>SMS2SESSID</code> . After idle time, overlay then redirect to <code>login.php?timeout=1</code> .	<code>config/session.php</code> : <code>HttpOnly</code> , <code>SameSite Lax</code> , <code>Secure</code> on <code>HTTPS</code> , <code>session.use_strict_mode</code> ; <code>smsEnforceSessionTimeout()</code> ; <code>assets/js/idle-logout.js</code>
Security headers + CSP	Browser DevTools → Network → response headers on any page.	<code>smsSendSecurityHeaders()</code> : <code>X-Frame-Options SAMEORIGIN</code> , <code>X-Content-Type-Options nosniff</code> , <code>CSP default-src self (+ Turnstile)</code>
Prepared SQL statements	Not a visible widget; identity lookups still use bound parameters.	<code>PDO prepare</code> / <code>execute</code> throughout CRAD helpers and sidebar queries
HTML escaping	User-supplied titles and names render as text, not HTML.	<code>e()</code> → <code>htmlspecialchars(..., ENT_QUOTES)</code>
Secure file uploads	Chapter / clearance upload accepts PDF/DOC/images only; files are not directly executable from the web.	<code>includes/uploads.php</code> : <code>MIME + extension whitelist</code> , random stored names, <code>storage/uploads/.htaccess</code> <code>Deny from all</code>
Secrets encryption at rest	System Settings stores Turnstile / SMTP secrets; they are not shown in plaintext after save.	<code>includes/crypto.php</code> <code>AES-256-GCM</code> ; <code>TOTP secrets</code> in <code>sms2_user_authenticators</code>
Activity audit log	Super Admin → Activity Logs; also Security Settings module logs. Failed logins are recorded.	<code>logActivity()</code> → <code>sms2_activity_logs</code> (user, role, action, IP, user agent)
Maintenance lock	Public users hit the maintenance page; admins use <code>?access=admin</code> .	<code>smsEnforceSystemMaintenance()</code> / <code>smsIsModuleInMaintenance()</code>
Title approval integrity	Group is not official until three statuses and three signatures exist (Section 3).	<code>cradValidTitleApprovalWhereSql()</code>
Grant sequential signatures	Approval Workflows: only the current office can YES/NO; NO returns the proposal.	<code>grantApprovalStepDefinitions()</code> and step-ownership checks
Password reset workflow	Staff request a reset; Super Admin approves in Module Security.	<code>sms2_password_reset_requests</code> (pending / approved / rejected)

9. Limitation to disclose

The student sidebar "Research Forum paid" lock still uses a hardcoded sample payment list in `includes/sidebar.php`, not a live ledger query. Do not claim that lock is production-complete unless it is wired to Payment History.